

The logo features a light blue square with a white, abstract, web-like pattern of intersecting lines. The text "VirtualJBoss" is overlaid on the left side of the square, with "Virtual:" in a smaller, lighter blue font and "JBoss" in a larger, bold, dark blue font.

VirtualJBoss

DOCKER AND JBOSS

THE PERFECT COMBINATION

[@marekgoldmann](#)

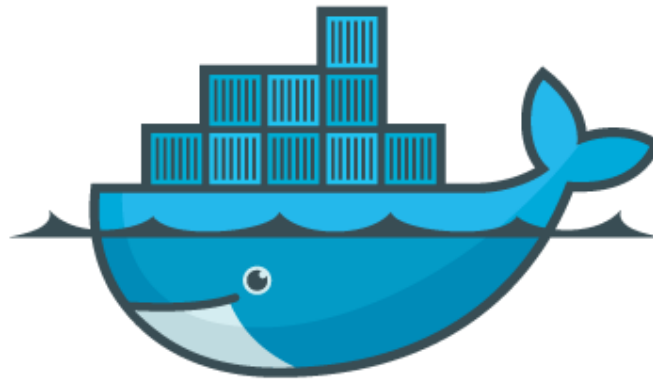
Virtual JBoss User Group, 9th Sep 2014

@MAREKGOLDMANN

- JBoss Software Engineer
 - Cloud-related stuff
 - Docker
- Fedora packager
 - `yum install wildfly`
 - `yum install docker-registry`

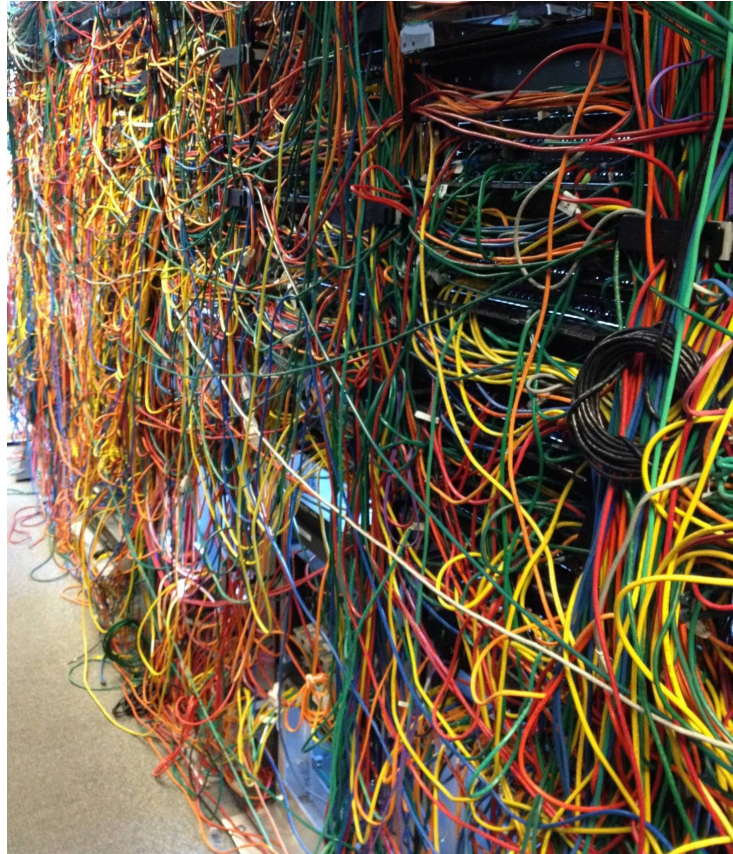
AGENDA

- **Small intro to Docker**
 - Why?
 - Comparison to virtual machines
- **JBoss and Docker**
 - Current state
- **Demos**
 - Basic demos
 - WildFly demos



docker

APPLICATIONS ARE COMPLEX

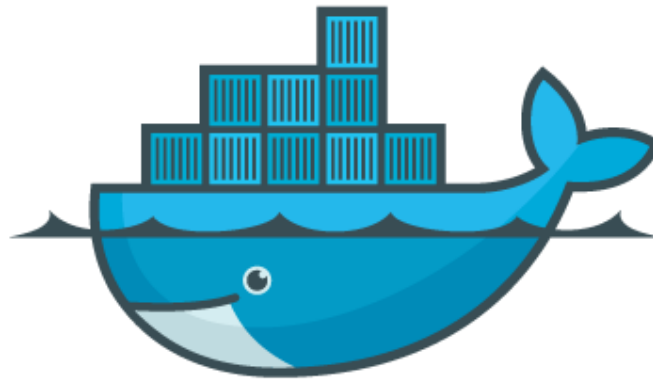


http://www.reddit.com/r/pics/comments/nf8la/im_always_tempted_to_switch_two_random_cables_to/

DIFFERENT DEPLOYMENT ENVIRONMENTS



<http://www.shedworking.co.uk/2012/03/nuts-and-bolts-cable-ties.html>



docker

A project to manage **containers**
...and **images**

CONTAINERS?

Yes, a lightweight operating system virtualization.

VIRTUAL MACHINES?

No, it's more lightweight.

VM (KVM, VMware)

We need to run the **whole OS** to start the process.

Container (LXC, Docker)

We launch processes **directly**, without booting the OS.

SO, HOW THIS COMPARES TO A
VIRTUAL MACHINE?

It's **completely** different.

VM VS. CONTAINER

	Virtual Machine (KVM, VMware)	Container (LXC, Docker)
Hardware	Simulated	Uses it (almost) directly
Supported OS'es	Almost any	Only Linux
Startup time	Seconds to minutes	Miliseconds
Scalability	A few	Sky is the limit (thousands)
Size	HUGE	Small

LINUX FTW!

[**https://github.com/boot2docker/boot2docker**](https://github.com/boot2docker/boot2docker)

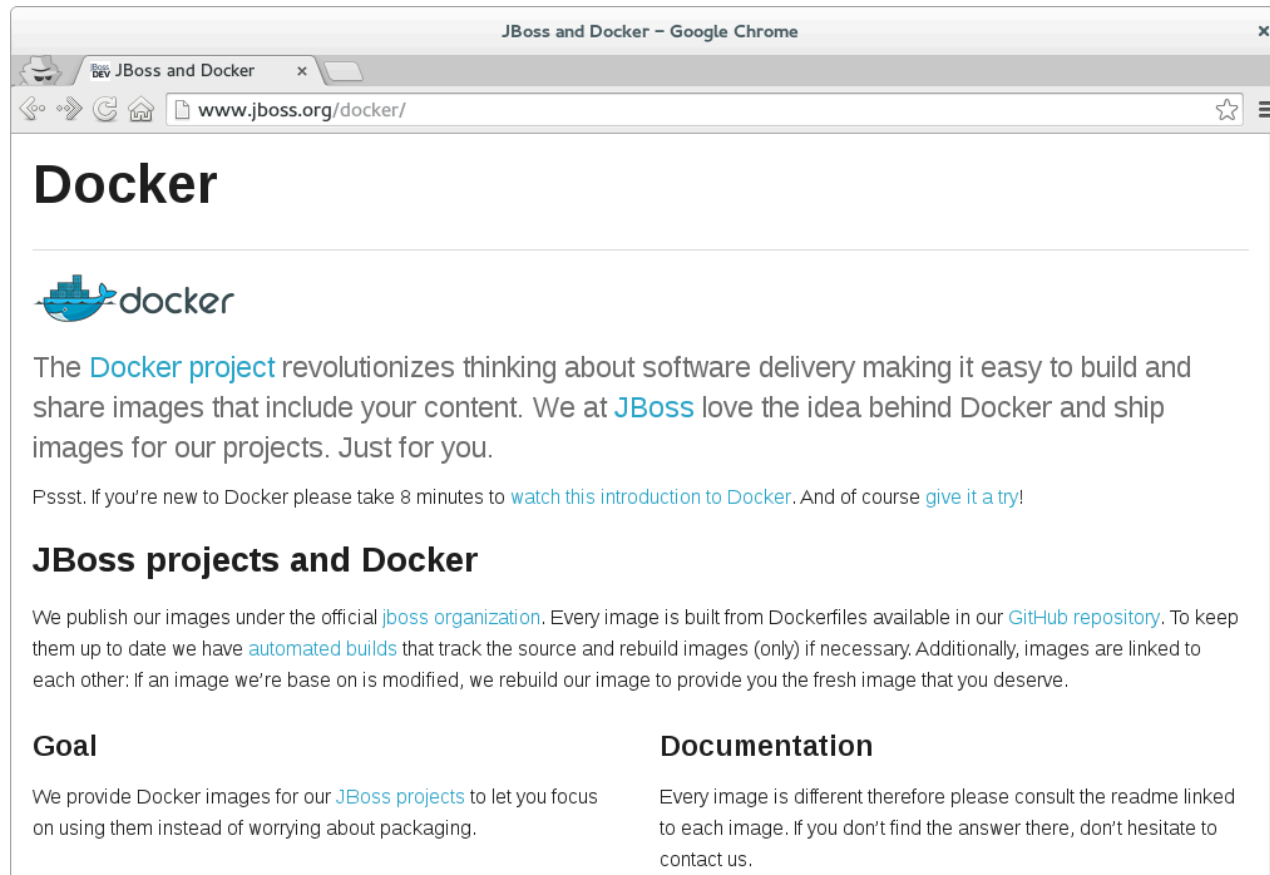
DOCKER COMMUNITY...

- **GitHub**
 - >1100 watchers
 - >14000 stars
 - >2700 forks
 - >660 open issues
 - >100 pull requests
- Over **585 contributors**
 - avg 17 commits / day
 - **90% are external**
- Many meetups all over the world
 - **50** active **meetup** groups

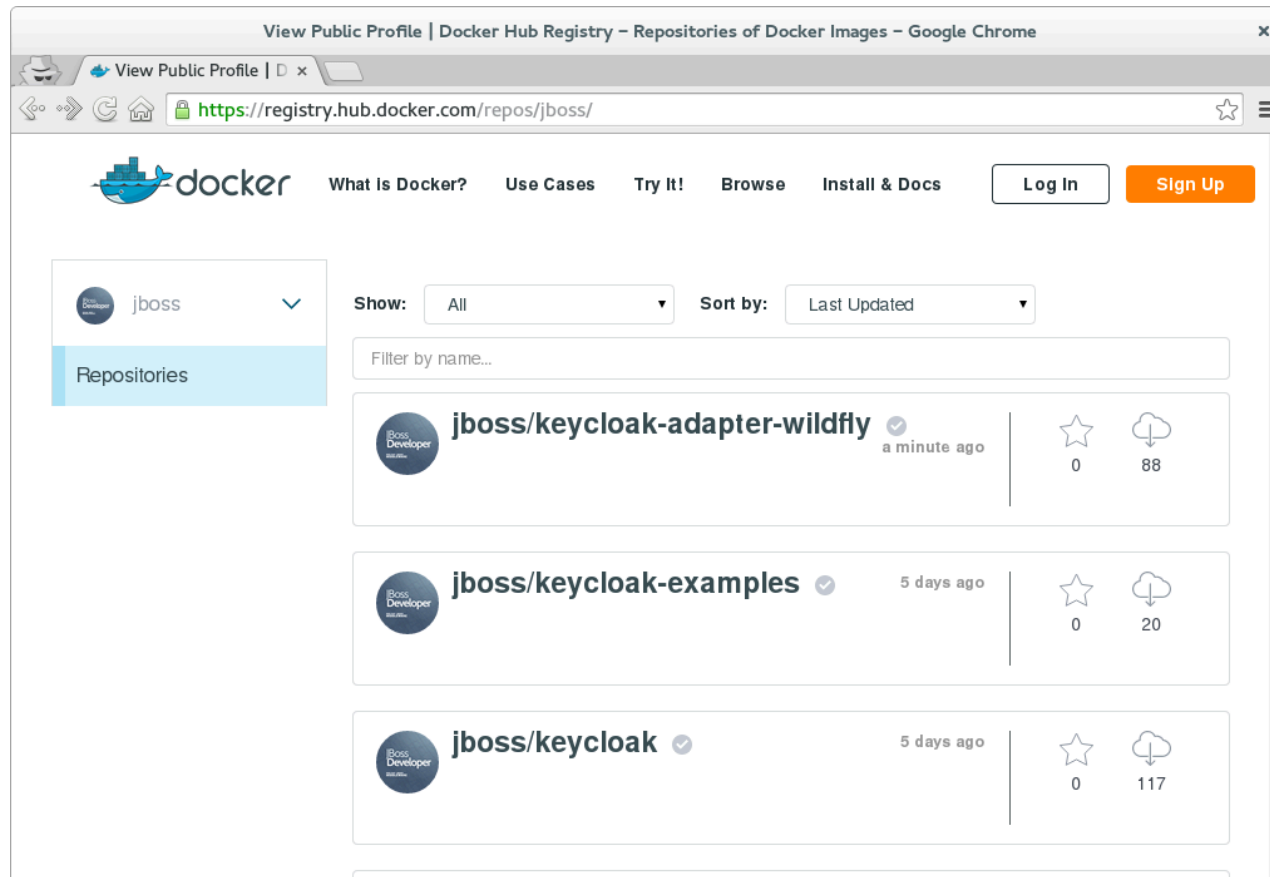
JBOSS AND DOCKER

DOCKERIZED PROJECTS

- WildFly
- Keycloak
- TorqueBox
- Aerogear
- Immutant
- Nodyn
- LiveOak
- ...



<http://www.jboss.org/docker/>



<https://registry.hub.docker.com/repos/jboss/>

GitHub page for **jboss/dockerfiles**. The repository is set of Dockerfiles for JBoss projects, located at <http://www.jboss.org/>.

Repository statistics: 54 commits, 1 branch, 0 releases, 10 contributors.

Current branch: **master**. The repository contains a directory **dockerfiles**.

Recent commits:

- Updated Keycloak to 1.0-rc-2 (latest commit a37863de12)
- aerogear**: Removing java and maven installations. 2 months ago
- immutant**: Bump immutant to 1.1.4. 5 days ago
- keycloak**: Updated Keycloak to 1.0-rc-2. 18 minutes ago
- liveoak**: Update to LiveOak Alpha02. 2 days ago
- nodyn**: Removing non required user and updating readme file. 20 days ago
- torquebox**: TorqueBox upgrade to 3.1.1. 2 months ago
- wildfly**: wildfly: Perform chown in same step as untar, to save 100Mb. 16 days ago
- LICENSE**: Updated README + added LICENSE file. 3 months ago

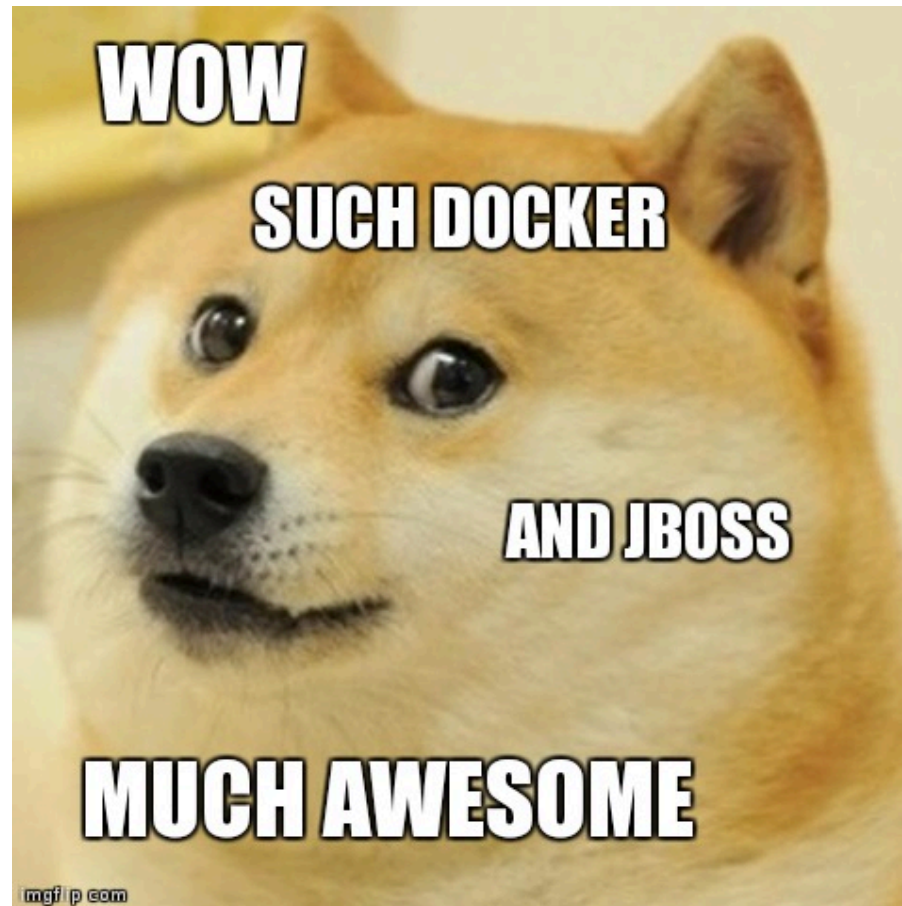
Right sidebar options: Code, Issues (6), Pull Requests (2), Pulse, Graphs.

HTTPS clone URL: <https://github.com/jboss/dockerfiles>

You can clone with [HTTPS](#) or [Subversion](#).

[Download ZIP](#)

<https://github.com/jboss/dockerfiles>



DEMOS!

A FEW TIPS

I NEED METRICS!

- Use iptables for network
- Parse /sys/fs/cgroup/ for everything else

I NEED TO LIMIT RESOURCES!

- CPU: `docker run -c 512 [...]`
- Memory: `docker run -m 128m [...]`
- Disk space: 10GB by default
 - **`docker -d --storage-opt dm.basesize=20G`**

I NEED TO DO BACKUP!

- Files?
 - Use volumes
- Data?
 - Use links

I NEED TO ACCESS MY CONTAINERS!

- Do you own the host?
 - Use nsenter
- In other cases:
 - Run sshd inside of the container (boooo...)

I NEED PERFORMANCE! RLY?

- CPU: native
- Memory: (almost) no overhead
- Network: no overhead with `--net host`
- I/O: native on volumes

THANKS!



Attributions:

<https://www.docker.io/>

